

3D Print Failure Diagnosis Guide

42 common failures — exact settings fixes — all materials

■ Heat & Clogs

Heat creep, jams, cold-end blockages, thermistor drift

■ Layer Issues

Delamination, gaps, under/over-extrusion, banding

■ Material Drying

Time & temp chart for PLA, PETG, PA, ABS, TPU, PC

■ Calibration

E-steps, flow rate, pressure advance, Z-offset

■ Adhesion

Warping, elephant foot, first layer lifting, stringing

■ Extruder/Motion

Skipping, grinding, belt tension, resonance tuning

Quick-Fix Reference

Symptom → Most likely cause → First fix to try

Symptom	Likely Cause	First Fix
Stringing	Retraction too low / temp too high	Raise retract 0.5mm, drop temp 5°C
Layer splits	Temp too low or speed too high	Raise nozzle temp 5°C, slow 10%
Warping	Bed temp / adhesion / cooling	Raise bed 5°C, add brim, kill drafts
Elephant foot	Z-offset too close	Raise Z-offset 0.02mm increments
Clog / grind	Heat creep or too-fast retract	Clean cold pull, check cooling fan
Extruder skip	Idler tension or high current	Tighten idler, reduce print speed
Gaps in top layer	Under-extrusion or low infill	Increase flow 2%, raise top layers
Z-banding	Lead screw wobble or resonance	Check coupler, tune input shaping
Ghosting/ringing	Speed too high, resonance	Enable input shaping / reduce accel
PA/Nylon warping	Wet filament + poor adhesion	Dry 70°C 6h, PVA glue stick bed
PETG stringing	Temp too high + retract too low	Drop to 230°C, retract 1mm direct
Under-extrusion PA	Moisture + inadequate temp	Dry filament, print 250°C+
Blob on start	Z-hop or ooze before prime	Wipe macro, purge line, Z-hop
First layer too thin	Z too close, flow too high	Raise Z-offset, tune live adjust

Filament Drying Reference

Material	Temp	Duration	Notes
PLA	45-50°C	4h	Often skippable if sealed
PETG	65°C	4-6h	Stringing drops dramatically
ABS	60-70°C	4h	Store sealed, prints better
PA/Nylon	70-80°C	8-12h	Critical — absorbs fast
TPU	50°C	4h	Reduces stringing & gaps
PC	80°C	6-8h	Always dry before use